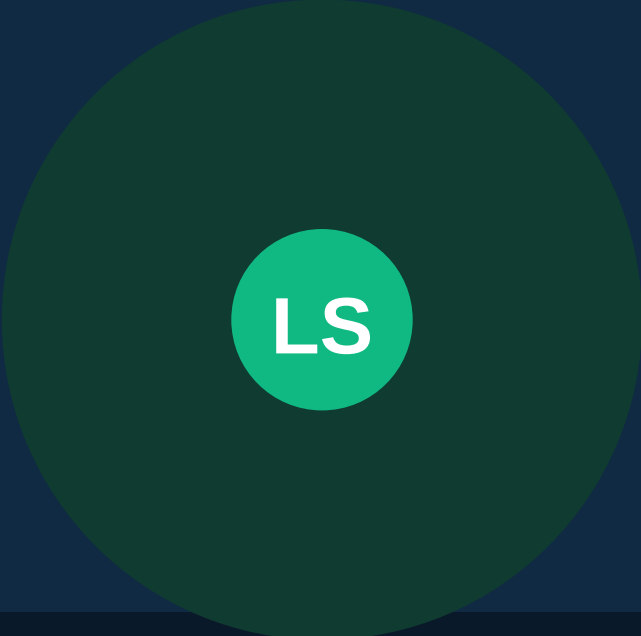




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LifeSync

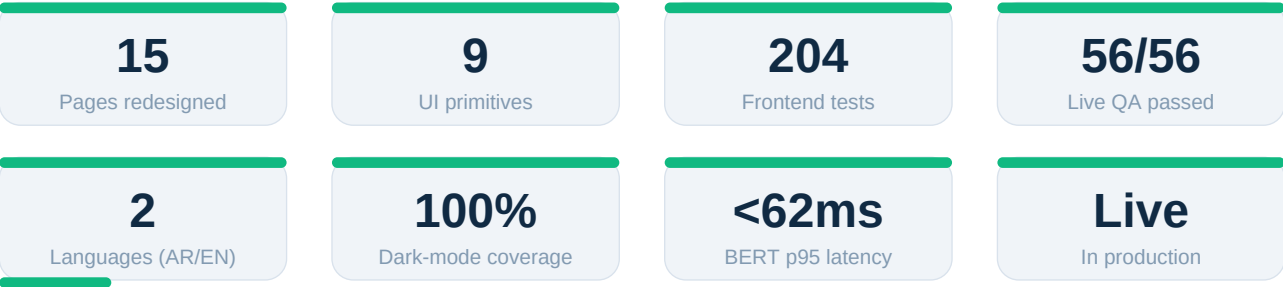
Your health and finances, one conversation away.

AI-powered life-management platform · Arabic-first, dark-mode, real-time voice

Project Portfolio & Engineering Showcase

Project Overview

LifeSync is a full-stack platform that unifies health and personal-finance tracking behind a single conversational AI. Users just describe their day in natural language — "walked 9,000 steps, spent \$45 on groceries" — and the system classifies, logs, and correlates the data across domains, surfacing insights a siloed app would miss. It ships fully bilingual (Arabic-first + English), with class-based dark mode and a real-time, hands-free voice assistant.



The Product in One Line

One dashboard for a complete life picture — health + finances visualized together, so you can see how spending habits track against sleep, mood, and activity, all driven by plain conversation.

Architecture & AI Routing

A two-track chat pipeline keeps intelligence deterministic where it matters and flexible where it helps:

Track A — BERT	A private, in-server BERT intent classifier (6 labels, ~62 ms p95) deterministically extracts entities and owns all data logging + cross-domain linking. Always runs; never hallucinates a log.
Track B — LLM	The user-selected model (routed through OpenRouter with a single key) writes the prose reply, with full history, memory, and just-logged facts as context. Switching models mid-chat changes only this track.
Voice	Real-time SSE token streaming feeds a sentence-by-sentence text-to-speech loop with Gemini-style barge-in — interrupting cancels TTS and aborts the in-flight model call, saving tokens.

Key Engineering Work

Design system	Built a 9-component UI library (Button, Card, Badge, Alert, FormField, Modal, StepWizard, FilterBar, Skeleton) with colocated tests, static Tailwind token maps, and a shared keyboard focus-ring.
Dark mode	Introduced a theme-invariant "ink" token scale to fix an inversion bug where decorative dark surfaces flipped near-white. Extended coverage to canvas charts via a chartTheme(isDark) helper — Chart.js paints to canvas, so CSS inversion can't reach it.

Arabic / RTL	Full RTL support via logical properties, browser-language auto-detection, 300+ i18n keys, and complete bilingual legal documents (Privacy + Terms) keyed by locale.
Interactive charts	Real cross-domain analytics: a D3 linked health/finance timeline with synced hover crosshair, plus Chart.js correlation, spending, and mood-vs-activity views — all theme-aware.
Chat streaming	Wired the parsed-but-unused SSE delta event into a ref-backed buffer flushed via requestAnimationFrame — live token streaming without a re-render per token.
QA & delivery	204 frontend unit tests, a Playwright API/UI suite, and a 56-check live black-box QA suite (security headers, CORS, auth contracts, BERT accuracy). Deployed frontend to Cloudflare Workers, backend on Railway.

Results & Verification

Every claim below was verified against the live production deployment, not just locally:

- 56 / 56 live QA checks pass — security, CORS, auth enforcement, error contracts, and raw BERT classifier accuracy (4/4 unambiguous intents matched at >0.98 confidence).
- 204 frontend tests green across 22 files; production build and lint clean at every step.
- Seeded 14 days of realistic health + finance data live; dashboard insights returned a computed health score of 73 with 4 detected cross-domain patterns.
- Deployed and smoke-verified: all SPA routes, Railway health, in-server BERT, OpenRouter, and voice config confirmed live.

Technology Stack

Frontend	React 19, Vite 7, Tailwind CSS 4 (@theme tokens), Chart.js + D3, custom i18n/RTL, Vitest + RTL.
Backend	Node/Express, Sequelize ORM, JWT auth + refresh rotation, bcrypt, SSE streaming, rate limiting.
AI	In-server BERT intent classifier (ONNX runtime) + OpenRouter-routed LLMs; deterministic dashboard engine.
Infra	Cloudflare Workers (frontend), Railway (backend + BERT), Firebase chat sync, GitHub Actions CI.

Live: lifesync.1202883.workers.dev · **API:** lifesync-production-fdf9.up.railway.app